

Repository query return types

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Supported Query Return Types

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The following table lists the return types generally supported by Spring Data repositories. However, consult the store-specific documentation for the exact list of supported return types, because some types listed here might not be supported in a particular store.

NOTE

Geospatial types (such as `GeoResult`, `GeoResults`, and `GeoPage`) are available only for data stores that support geospatial queries. Some store modules may define their own result wrapper types.

Table 1. Query return types

Return type	Description
<code>void</code>	Denotes no return value.
Primitives	Java primitives.
Wrapper types	Java wrapper types.
<code>T</code>	A unique entity. Expects the query method to return one result at most. If no result is found, <code>null</code> is returned. More than one result triggers an <code>IncorrectResultSizeDataAccessException</code> .
<code>Iterator<T></code>	An <code>Iterator</code> .
<code>Collection<T></code>	A <code>Collection</code> .
<code>List<T></code>	A <code>List</code> .
<code>Optional<T></code>	A Java 8 or Guava <code>Optional</code> . Expects the query method to return one result at most. If no result is found, <code>Optional.empty()</code> or <code>Optional.absent()</code> is returned. More than one result triggers an <code>IncorrectResultSizeDataAccessException</code> .

Return type	Description
<code>Option<T></code>	Either a Scala or Vavr <code>Option</code> type. Semantically the same behavior as Java 8's <code>Optional</code> , described earlier.
<code>Stream<T></code>	A Java 8 <code>Stream</code> .
<code>Streamable<T></code>	A convenience extension of <code>Iterable</code> that directly exposes methods to stream, map and filter results, concatenate them etc.
Types that implement <code>Streamable</code> and take a <code>Streamable</code> constructor or factory method argument	Types that expose a constructor or <code>...of(...)</code> / <code>...valueOf(...)</code> factory method taking a <code>Streamable</code> as argument. See Returning Custom Streamable Wrapper Types for details.
Vavr <code>Seq</code> , <code>List</code> , <code>Map</code> , <code>Set</code>	Vavr collection types. See Support for Vavr Collections for details.
<code>Future<T></code>	A <code>Future</code> . Expects a method to be annotated with <code>@Async</code> and requires Spring's asynchronous method execution capability to be enabled.
<code>CompletableFuture<T></code>	A Java 8 <code>CompletableFuture</code> . Expects a method to be annotated with <code>@Async</code> and requires Spring's asynchronous method execution capability to be enabled.
<code>Slice<T></code>	A sized chunk of data with an indication of whether there is more data available. Requires a <code>Pageable</code> method parameter.
<code>Page<T></code>	A <code>Slice</code> with additional information, such as the total number of results. Requires a <code>Pageable</code> method parameter.
<code>Window<T></code>	A <code>Window</code> of results obtained from a scroll query. Provides <code>ScrollPosition</code> to issue the next scroll query. Requires a <code>ScrollPosition</code> method parameter.
<code>GeoResult<T></code>	A result entry with additional information, such as the distance to a reference location.
<code>GeoResults<T></code>	A list of <code>GeoResult<T></code> with additional information, such as the average distance to a reference location.
<code>GeoPage<T></code>	A <code>Page</code> with <code>GeoResult<T></code> , such as the average distance to a reference location.
<code>SearchResult<T></code>	A result entry with additional information, such as the score in relation to the reference search term.
<code>SearchResults<T></code>	A list of <code>SearchResult<T></code> .
<code>Mono<T></code>	A Project Reactor <code>Mono</code> emitting zero or one element using reactive repositories. Expects the query method to return one result at most. If no result is found, <code>Mono.empty()</code> is returned. More than one result triggers an <code>IncorrectResultSizeDataAccessException</code> .

Return type	Description
<code>Flux<T></code>	A Project Reactor <code>Flux</code> emitting zero, one, or many elements using reactive repositories. Queries returning <code>Flux</code> can emit also an infinite number of elements.
<code>Single<T></code>	A RxJava <code>Single</code> emitting a single element using reactive repositories. Expects the query method to return one result at most. If no result is found, <code>Mono.empty()</code> is returned. More than one result triggers an <code>IncorrectResultSizeDataAccessException</code> .
<code>Maybe<T></code>	A RxJava <code>Maybe</code> emitting zero or one element using reactive repositories. Expects the query method to return one result at most. If no result is found, <code>Mono.empty()</code> is returned. More than one result triggers an <code>IncorrectResultSizeDataAccessException</code> .
<code>Flowable<T></code>	A RxJava <code>Flowable</code> emitting zero, one, or many elements using reactive repositories. Queries returning <code>Flowable</code> can emit also an infinite number of elements.



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